Elementary central binomial coefficient estimates

Asked 11 years, 1 month ago Modified 1 year, 7 months ago Viewed 9k times

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1. How to prove that $\frac{4^n}{\sqrt{4n}} < \binom{2n}{n} < \frac{4^n}{\sqrt{3n+1}}$ for all $n > 1$?

2. Does anyone know any better elementary estimates?

Attempt. We have

$$\frac{1}{2^n}\binom{2n}{n} = \prod_{k=0}^{n-1} \frac{2n-k}{2(n-k)} = \prod_{k=0}^{n-1} \left(1 + \frac{k}{2(n-k)}\right).$$

Then we have

$$\left(1 + \frac{k}{2(n-k)}\right) > \sqrt{1 + \frac{k}{n-k}} = \frac{\sqrt{n}}{\sqrt{n-k}}.$$

So maybe, for the lower bound, we have

$$\frac{n^{\frac{n}{2}}}{\sqrt{n!}} = \prod_{k=0}^{n-1} \frac{\sqrt{n}}{\sqrt{n-k}} > \frac{2^n}{\sqrt{4n}}.$$

By Stirling, $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, so the lhs becomes

$$\frac{e^{\frac{n}{2}}}{(2\pi n)^{\frac{1}{4}}},$$

but this isn't $> \frac{2^n}{\sqrt{4n}}$.

discrete-mathematics inequality binomial-coefficients approximation

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edited Jul 15, 2020 at 13:32
Batominovski
48.6k 4 51 125

asked Aug 19, 2011 at 20:26
Robert William Hanks
789 7 9

2 For (2), Stirling's approximation yields $\binom{2n}{n} \sim (\pi n)^{-1/2} 4^n$, and you might look at some of the other formulas for sharper bounds. – anon Aug 19, 2011 at 20:49

1 @anon The Central Limit Theorem shows that eventually the Stirling approximation must be an upper bound (and it actually is for all positive n). A lower bound is obtained the same way by

multiplying the upper bound by $\exp(-1/(4n))$. Because this is $1+O(1/n)$ for large n, it's better than the OP's bounds whose ratio equals $O(\sqrt{4/3})$. – whuber Aug 19, 2011 at 20:55

en.wikipedia.org/wiki/Central_binomial_coefficient – yoyo Aug 19, 2011 at 21:06

1 @whuber: 1,000,000=O(sqrt(4/3)). en.wikipedia.org/wiki/Big_O_notation#Formal_definition – Did Aug 20, 2011 at 12:32

I believe that you can find better estimates in the papers "Tikhonov, I. V.; Sherstyukov, V. B.; Tsvetkovich, D. G. Comparative analysis of two-sided estimates of the central binomial coefficient. Chelyab. Fiz.-Mat. Zh. 5 (2020), no. 1, 70-95; available online at doi.org/10.24411/2500-0101-2020-15106." and "Popov, A. Yu. Two-sided estimates of the central binomial coefficient. Chelyab. Fiz.-Mat. Zh. 5 (2020), no. 1, 56-69; available online at doi.org/10.24411/2500-0101-2020-15105." – qifeng618 Sep 14, 2021 at 4:43

7 Answers

Sorted by: Highest score (default)

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Here are some crude bounds:

$$\frac{1}{2\sqrt{n}} \le \binom{2n}{n}\frac{1}{2^{2n}} \le \frac{3}{4\sqrt{n+1}}, \quad n \ge 1.$$

We begin with the product representations

$$\binom{2n}{n}\frac{1}{2^{2n}} = \frac{1}{2n}\prod_{j=1}^{n-1}\left(1+\frac{1}{2j}\right) = \prod_{j=1}^{n}\left(1-\frac{1}{2j}\right), \quad n \ge 1.$$

From

$$\prod_{j=1}^{n-1}\left(1+\frac{1}{2j}\right)^2 = \prod_{j=1}^{n-1}\left(1+\frac{1}{j}+\frac{1}{4j^2}\right) \ge \prod_{j=1}^{n-1}\left(1+\frac{1}{j}\right) = n,$$

we see that

$$\left(\binom{2n}{n}\frac{1}{2^{2n}}\right)^2 = \frac{1}{(2n)^2}\prod_{j=1}^{n-1}\left(1+\frac{1}{2j}\right)^2 \ge \frac{1}{4n^2}n = \frac{1}{4n}, \quad n \ge 1.$$

so by taking square roots, $\binom{2n}{n}\frac{1}{2^{2n}} \ge \frac{1}{2\sqrt{n}}$.

On the other hand,

$$\prod_{j=1}^{n}\left(1+\frac{1}{2j}\right)\left(1-\frac{1}{2j}\right) = \prod_{j=1}^{n}\left(1-\frac{1}{4j^2}\right) \le \frac{3}{4},$$

so that (using the lower bound above), we have

$$\binom{2n}{n}\frac{1}{2^{2n}} = \prod_{j=1}^{n}\left(1-\frac{1}{2j}\right) \leq \frac{3}{4\sqrt{n+1}}.$$

Alternatively, multiplying the different representations we get

$$n\left[\binom{2n}{n}\frac{1}{2^{2n}}\right]^2 = \frac{1}{2}\prod_{j=1}^{n-1}\left(1-\frac{1}{4j^2}\right)\left(1-\frac{1}{2n}\right).$$

It's not hard to show that the right hand side increases from $1/4$ to $1/\pi$ for $n \geq 1$.

Edit: You can get better bounds if you know Wallis's formula:

$$2n\left[\binom{2n}{n}\frac{1}{4^n}\right]^2 = \frac{1}{2}\frac{3}{2}\frac{3}{4}\frac{5}{4}\cdots\frac{2n-1}{2n-2}\frac{2n-1}{2n} = \frac{1}{2}\prod_{j=2}^{n}\left(1+\frac{1}{4j(j-1)}\right)$$

$$(2n+1)\left[\binom{2n}{n}\frac{1}{4^n}\right]^2 = \frac{1}{2}\frac{3}{2}\frac{3}{4}\frac{5}{4}\cdots\frac{2n-1}{2n-2}\frac{2n-1}{2n}\frac{2n+1}{2n} = \prod_{j=1}^{n}\left(1-\frac{1}{4j^2}\right)$$

By Wallis's formula, both middle expressions converge to $\frac{2}{\pi}$. The right hand side of the first equation is increasing, while the right hand side of the second equation is decreasing. We conclude that

$$\frac{1}{\sqrt{\pi(n+1/2)}} \leq \binom{2n}{n}\frac{1}{4^n} \leq \frac{1}{\sqrt{\pi n}}.$$

Share Cite Edit Follow Flag edited May 29, 2012 at 12:10 answered Aug 19, 2011 at 20:53 user940

27

For $n \geq 0$, we have (by cross-multiplication)

$$\begin{aligned}\left(\frac{n+\frac{1}{2}}{n+1}\right)^2 &= \frac{n^2+n+\frac{1}{4}}{n^2+2n+1}\\ &\leq \frac{n+\frac{1}{3}}{n+\frac{4}{3}} \qquad (1)\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{\binom{2n+2}{n+1}}{\binom{2n}{n}} &= 4\frac{n+\frac{1}{2}}{n+1}\\ &\leq 4\sqrt{\frac{n+\frac{1}{3}}{n+\frac{4}{3}}} \qquad (2)\end{aligned}$$

Inequality (2) implies that

$$\boxed{\binom{2n}{n}\frac{\sqrt{n+\frac13}}{4^n}\text{ is decreasing}}\tag{3}$$

For $n\ge0$, we have (by cross-multiplication)

$$\begin{aligned}\left(\frac{n+\frac12}{n+1}\right)^2&=\frac{n^2+n+\frac14}{n^2+2n+1}\\&\ge\frac{n+\frac14}{n+\frac54}\end{aligned}\tag{4}$$

Therefore,

$$\begin{aligned}\frac{\binom{2n+2}{n+1}}{\binom{2n}{n}}&=4\frac{n+\frac12}{n+1}\\&\ge4\sqrt{\frac{n+\frac14}{n+\frac54}}\end{aligned}\tag{5}$$

Inequality (5) implies that

$$\boxed{\binom{2n}{n}\frac{\sqrt{n+\frac14}}{4^n}\text{ is increasing}}\tag{6}$$

Note that the formula in (3), which is decreasing, is bigger than the formula in (6), which is increasing. Their ratio tends to 1; therefore, they tend to a common limit, L.

Theorem 1 from this answer says

$$\lim_{n\to\infty}\frac{\sqrt{\pi n}}{4^n}\binom{2n}{n}=1\tag{7}$$

which means that

$$\begin{aligned}L&=\lim_{n\to\infty}\frac{\sqrt{n}}{4^n}\binom{2n}{n}\\&=\frac{1}{\sqrt{\pi}}\end{aligned}\tag{8}$$

Combining (3), (6), and (8), we get

$$\boxed{\frac{4^n}{\sqrt{\pi\left(n+\frac13\right)}}\le\binom{2n}{n}\le\frac{4^n}{\sqrt{\pi\left(n+\frac14\right)}}}\tag{9}$$

Assymptotic Observation

If we know that $n \geq n_0$, we can replace $\frac{1}{3}$ in $(1) - (3)$ and (9) by $\frac{1}{4} + \frac{1}{16n_0+12}$. Thus, asymptotically,

$$\boxed{\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi\left(n+\frac{1}{4}\right)}}} \tag{10}$$

Share Cite Edit Follow Flag edited Oct 7, 2019 at 4:09 answered Sep 15, 2014 at 16:22



robjohn ♦
330k 34 431 806

@ByronSchmuland: Thanks! I came up with the idea of the estimates (3) and (6) while working on this answer, then expanded on it when I found this question. – robjohn ♦ Sep 16, 2014 at 15:05

For finite n, you can use $0 < O(\frac{1}{n^2}) < \frac{1}{72n}$ in (11) (per Wikipedia). – A.S. Feb 23, 2016 at 17:36

@robjohn: +1 as far as I know, (10) gives the sharpest upper and lower bounds. Is there a reference so we can cite? Or you may write a note and put it in arXiv.org so we can site. – Shiyu Mar 1, 2016 at 19:38

@robjohn, I'd appreciate a reference to cite too. +1 – MikeY Dec 6, 2017 at 14:40

@MikeY: unfortunately, this was all done for this question. I don't have any other reference other than this answer. You can get a citation from the "cite" option in the same area as the "share" and "flag" options, below the answer. – robjohn ♦ Dec 6, 2017 at 14:57

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10

You can get an even more precise answer than those already provided by using more terms in the Stirling series. Doing so yields, to a relative error of $O(n^{-5})$,

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}}\left(1 - \frac{1}{8n} + \frac{1}{128n^2} + \frac{5}{1024n^3} - \frac{21}{32768n^4} + O(n^{-5})\right).$$

To the same relative error of $O(n^{-5})$, the Stirling series is

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n\left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} - \frac{571}{2488320n^4} + O(n^{-5})\right).$$

Then we have

$$\binom{2n}{n} = \frac{(2n)!}{n!n!} = \frac{4^n}{\sqrt{\pi n}}\frac{1 + \frac{1}{12(2n)} + \frac{1}{288(2n)^2} - \frac{139}{51840(2n)^3} - \frac{571}{2488320(2n)^4} + O(n^{-5})}{\left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} - \frac{571}{2488320n^4} + O(n^{-5})\right)^2}.$$

Crunching through the long division with the polynomial in $\frac{1}{n}$ (which Mathematica can do immediately with the command `Series[expression, {n, ∞, 4}]`) yields the expression for $\binom{2n}{n}$ at the top of the post.

See also Problem 9.60 in *Concrete Mathematics* (2nd edition).

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answered Aug 23, 2011 at 16:52



Mike Spivey

53.4k 17 169 273

$\frac{4^n}{\sqrt{\pi n}}\left(1-\frac{1}{8n}+O\left(\frac{1}{n^2}\right)\right)$ matches the upper bound gotten in my answer. I remember computing this asymptotic series at one point. (+1) – robjohn ♦ Sep 16, 2014 at 10:38

9

A way to get explicit bounds via Stirling's approximation is to use the following more precise formulation:

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\alpha_n}$$

where $\frac{1}{12n+1} < \alpha_n < \frac{1}{12n}$.

With this one arrives at

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} e^{\lambda_n}$$

where $\frac{1}{24n+1} - \frac{1}{6n} < \lambda_n < \frac{1}{24n} - \frac{2}{12n+1}$.

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edited Mar 14, 2013 at 0:21

answered Aug 20, 2011 at 13:12



Ragib Zaman

33.8k 3 66 109

@ThomasAndrews Ahh yes thank you for pointing that out. – Ragib Zaman Mar 14, 2013 at 0:21

1

$\binom{2p}{p}$ central binomial coefficient

$$\binom{2p}{p} = \frac{2^{2p}}{1+\sum_{n=1}^{p}\frac{2^n}{S}}$$

with $S = \sum_{k=1}^{2n}\left(1+\cos\left(\frac{k\pi}{n}\right)\right)^n$

this new formula as the first uses a trigonometric form (see mar20 at19:11) it's more complex but gives the same result

with my pocket computer casio pb 700 I have found $\binom{2p}{p}$ exact untill

$p = 18$ it's 9075135300 after the result is given $A \cdot 10^x$

example $\binom{40}{20} = 1.378465288 \times 10^{11}$

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edited Mar 31, 2016 at 9:36
robjohn ♦
330k 34 431 806

answered Mar 31, 2016 at 9:16
babois jean-claude
11 1

I have applied MathJax. Please make sure everything is as intended. MathJax makes things easier to read. – robjohn ♦ Mar 31, 2016 at 9:38

1

Here is a better estimate of the quantity. I am not going to prove it. Since I know about it, hence I am sharing and will put the reference.

$$\frac{1}{2\sqrt{n}} \leq \binom{2n}{n} 2^{-2n} \leq \frac{1}{\sqrt{2n}}$$

Refer to Page 590 This is the text book "Boolean functions Complexity" by Stasys Jukna.

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answered Mar 1, 2021 at 4:38
Amit
131 6

Beside the fact that link-only answers are discouraged because the links can go stale, the linked PDF provides no hint of how to prove this inequality either; it simply provides a list of many formulas and inequalities with no proof. In any case, this inequality is less precise than the inequality given in the question. – robjohn ♦ Mar 1, 2021 at 7:24

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$$\binom{2n}{n} = \frac{2^{n-1}}{n} \sum_{k=0}^{2n} \left(1 + \cos\frac{k\pi}{n}\right)^n$$

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edited Mar 20, 2016 at 19:11
Alex M.
34.2k 15 44 83

answered Mar 20, 2016 at 17:35
babois jean-claude
11 1

And how did you get that formula? – Konstantinos Gaitanas Dec 30, 2021 at 7:04